

Writing the Inverse of a Linear Function

TEACHER KEY | Algebra II Honors | Unit 1 | Day 11 | TEK A2.2.B

Objective: I can write the inverse of a linear function algebraically, derive the inverse of the general form $f(x) = mx + b$, and prove that a line and its inverse have reciprocal slopes.

Warm-Up — This Is a Skill You Have

Finding an inverse is solving a literal equation for the other variable. Nothing new.

Solve $y = 3x + 6$ for x : $x = (y - 6)/3$ Solve $y = mx + b$ for x : $x = (y - b)/m$

The Procedure

1. Replace $f(x)$ with y .
2. **Swap x and y .**
3. Solve for y and rename it **$f^{-1}(x)$.**

Part 1 — Find Each Inverse

Show the swap-and-solve steps, not just the answer.

a. $f(x) = 2x + 3$

$$f^{-1}(x) = (x - 3)/2$$

b. $f(x) = -2x + 8$

$$f^{-1}(x) = (8 - x)/2$$

c. $f(x) = (1/2)x - 6$

$$f^{-1}(x) = 2x + 12$$

d. $f(x) = (3x - 1)/4$

$$f^{-1}(x) = (4x + 1)/3$$

Part 2 — Derive the General Case

Now do it once for every linear function at the same time. Let $f(x) = mx + b$, with $m \neq 0$.

$$f^{-1}(x) = (x - b)/m \text{ which in slope-intercept form is } y = (1/m)x - b/m$$

Slope of f is **m** and slope of f^{-1} is **$1/m$** , so the slopes are **reciprocals**. That is now proved, not just noticed.

Why did the derivation require $m \neq 0$? What kind of line is $m = 0$, and what goes wrong?

A **horizontal** line. Every input maps to one output, so the inverse **fails the vertical line test**.

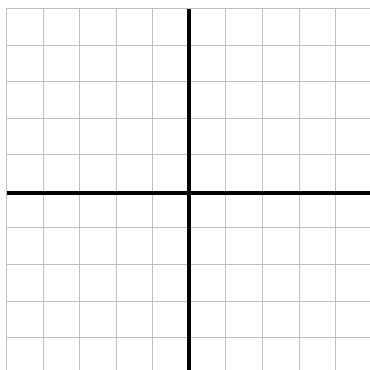
Part 3 — Check Two Ways

Use $f(x) = 2x + 3$ and $f^{-1}(x) = (x - 3)/2$.

x	-1	0	1	2
f(x)	1	3	5	7

x	1	3	5	7
f⁻¹(x)	-1	0	1	2

Graph f , f^{-1} , and $y = x$.



Record it.

slope of f **2** slope of f^{-1} **1/2**

y-int of f **3** of f^{-1} **-3/2**

Where do f and f^{-1} intersect?

on the line $y = x$, at $(-3, -3)$

Part 4 — Self-Inverse Lines

A function is self-inverse when $f^{-1}(x) = f(x)$. Using $f(x) = mx + b$, find every linear function with this property.

Need $(x - b)/m = mx + b$, which forces $m =$ **1 or -1**

If $m = 1$, b must be **0**, giving **$f(x) = x$** . If $m = -1$, b may be **anything**, giving $f(x) = -x + b$.

Check one: is $f(x) = -x + 6$ its own inverse? Test with $x = 2$.

$f(2) =$ **4** $f(4) =$ **2** so yes, it returns the start. **Self-inverse**

Part 5 — In Context

Celsius to Fahrenheit: $F(c) = (9/5)c + 32$.

$F^{-1}(x) =$ **$(5/9)(x - 32)$** In words, the inverse **converts Fahrenheit to Celsius**

Is there a temperature where the two scales agree? Set $(9/5)c + 32 = c$ and solve.

$c =$ **-40** — **$-40^{\circ}\text{C} = -40^{\circ}\text{F}$, the fixed point on $y = x$**

Exit Ticket

Write the inverse of $f(x) = 6x - 12$, verify with one input, and state the relationship between the two slopes.

$f^{-1}(x) =$ **$(x + 12)/6$** $f(3) =$ **6** and $f^{-1}(6) =$ **3**

Slopes: **6 and 1/6 - reciprocals**